

Repurposing natural gas pipelines for hydrogen: Limits and options from a case study in Germany

Kornél Télessy^{a,b}, Lukas Barner^{a,b}, Franziska Holz^{a,c,*}

^a German Institute for Economic Research (DIW Berlin), Mohrenstr. 58, 10117, Berlin, Germany

^b Technische Universität Berlin, Str. des 17. Juni 135, 10623, Berlin, Germany

^c Norwegian University of Science and Technology (NTNU), NTNU Energy Transition Initiative (NETI), Trondheim, Norway

ARTICLE INFO

Handling Editor: Suleyman I. Allakhverdiev

Keywords:

Hydrogen pipelines

Pipeline repurposing

Long term costs of transportation

Repurposing options

ABSTRACT

We investigate the challenges and options for repurposing existing natural gas pipelines for hydrogen transportation. Challenges of re-purposing are mainly related to safety and due to the risk of hydrogen embrittlement of pipeline steels and the smaller molecular size of the gas. From an economic perspective, the lower volumetric energy density of hydrogen compared to natural gas is a challenge. We investigate three pipeline repurposing options in depth: a) no modification to the pipeline but enhanced maintenance, b) use of gaseous inhibitors, and c) the pipe-in-pipe approach. The levelized costs of transportation of these options are compared for the case of the German Norddeutsche Erdgasleitung (NEL) pipeline. We find a similar cost range for all three options. This indicates that other criteria such as the sunk costs, public acceptance, and consumer requirements are likely to shape the decision making for gas pipeline repurposing.

1. Introduction

The effects of global warming have an increasing impact on our everyday life and the limitation of these impacts is one of the major challenges of our generation. The European Union has strengthened its climate targets in order to reach climate neutrality, i.e. zero net emissions, by 2050. To reach this goal, the use of fossil fuels such as natural gas needs to be phased out and the use of renewable energies must be substantially expanded (e.g. [1]). Hydrogen has an important role to play in an electrified energy system, e.g. as long-term storage option [2], as gaseous renewable fuel for high-temperature industrial applications and as input fuel for liquid e-fuels, e.g. for aviation [3].

Yet, there are significant challenges associated with the development of a new (green) hydrogen sector and one of them is the creation of hydrogen transportation infrastructure. Following reports like the European Hydrogen Backbone studies [4,5], there is an increasing perception that the bulk of future hydrogen supplies will be transported in pipelines, similar to the organization of the natural gas sector today. Given that natural gas infrastructure might become obsolete during its phase-out, the repurposing of pipelines has become an attractive idea to avoid stranding of fossil infrastructure while potentially reducing

investment costs of the new hydrogen pipeline network. At the European level, some actors, such as infrastructure regulators, started to obtain an overview of the technical possibilities and challenges (e.g. Ref. [6]), but no consensus on the best technical solution for pipeline repurposing has emerged yet.

This paper investigates options to convert a high-pressure natural gas pipeline to the transportation of hydrogen from an economic perspective. It also analyzes which critical elements must be considered for such an assessment. We present an overview of the most relevant pipeline conversion options, namely a) enhanced maintenance of an otherwise unchanged pipeline, b) the use of gaseous inhibitors, and c) the pipe-in-pipe approach. We discuss the technical challenges and costs associated with each option for relevant pipeline infrastructure items. Based on this data collection, a cost comparison is applied to the NEL¹ pipeline in Germany. While we do not believe that the NEL will be repurposed very soon, it is determined a particularly suitable case study. On the one hand, it is a rather new pipeline, built with state-of-the-art materials and technology. Data on these characteristics is publicly available.² On the other hand, the pipeline only consists of one entry inlet and one exit point, without additional branches. This makes the calculation of gas flows simpler than in the case of interconnected pipelines. We find that

* Corresponding author. German Institute for Economic Research (DIW Berlin), Mohrenstr. 58, 10117, Berlin, Germany.

E-mail address: franziska.holz@ntnu.no (F. Holz).

¹ NEL: North-German natural gas pipeline ("Norddeutsche Erdgasleitung").

² [12] has to make assumptions on the material of nine out of the 13 major pipelines in Germany.

Table 1
Properties of methane and hydrogen at standard temperature and pressure.

Parameter	Unit	CH ₄	H ₂	Source
Molecular weight	kg/mol	0.016043	0.002016	(1)
Density	kg/m ³	0.668160	0.083752	(1)
Lower heating value (gravimetric)	kWh/kg	13.9	33.3	(2)
Lower heating value (volumetric)	kWh/m ³	9.3	2.8	(3)
Diffusion coefficient in air	cm ² /s	0.16	0.61	(4)
Flammability limits in air	vol%	5.3–15	4.0–75	(4)
Minimum energy for ignition in air	mJ	0.29	0.02	(4)

Sources: (1): [14], (2): [15], (3): own calculation (density x gravimetric LHV), (4): [16].

all three considered repurposing options have similar leveled costs of hydrogen transport. The costs are also hardly sensitive to electricity price assumptions.

Our analysis updates and complements a number of studies published on the topic of repurposing existing natural gas infrastructure for hydrogen. Some studies, such as [7,8] consider blending of only a share of hydrogen into the gas stream. In contrast, we focus on a full conversion of the gas pipeline to 100% hydrogen use. [9] make a material fatigue analysis of gas pipelines if used for hydrogen transport while giving a brief overview of mitigation options for this problem. Similarly [10], provide an overview of the characteristics and challenges of hydrogen transportation in gas pipelines in which they emphasize the risks of hydrogen embrittlement and leakage but they ignore economic aspects such as costs. The European Hydrogen Backbone studies [4,5] present the possible extent of a European hydrogen transportation network, consisting to a large part of repurposed natural gas pipelines. At the same time, cost estimated for the leveled costs of transportation are presented. They compare to higher cost estimates for new-built (e.g. [11]).

However, except for Cerniauskas et al. [12], all studies are unclear about the concrete repurposing measures. This particular study [12] carries out an in-depth analysis of some of the same repurposing options as our study. However, they consider a centralized purification for added inhibitors and refrain from considering the pipe-in-pipe approach as a relevant repurposing option, as they assume the excavation of existing pipelines to be necessary. In our study, we use different assumptions than Cerniauskas et al. [12] which are discussed in more detail later. Finally, economic literature with a system-focused perspective, concentrate on determining the extent of a cost-optimal H₂ pipeline network but do not cover the technical issues in-depth (e.g. [13]).

Our study provides an extensive overview of the techno-economic challenges of gas pipeline repurposing for hydrogen transportation. In addition, we provide a transparent calculation of the leveled costs of transportation for the selected repurposing options in the case of the NEL pipeline. The next section presents the technical background of gaseous pipeline transportation (Section 2). Afterwards, we investigate the criteria that must be met by a pipeline that transports hydrogen (Section 3). In Section 4, we apply the three options of pipeline repurposing to the NEL pipeline in Germany and calculate the related costs. We discuss our findings and conclude in Section 5.

2. Technical background

2.1. Comparison of natural gas and hydrogen

Natural gas (NG) and hydrogen (H₂) differ fundamentally in their physical properties. These differences are the root cause of the problems

emerging within the repurposing of existing transportation infrastructure. For reasons of simplicity, the properties of methane (CH₄) are used when highlighting the differences of NG to H₂ and modelling pipeline operation, as methane is the main component of NG.³ An overview of the relevant properties of CH₄ and H₂ for transportation and their use as an energy carrier at standard temperature and pressure (STP, 293.15 K and 1 atm) is given in Table 1.

Being the smallest element in the periodic system, H₂ is far lighter than methane. This difference is one cause of many concerns, as it enables H₂ to diffuse easier and makes handling the gas more difficult. Known safety concerns are also reflected in Table 1 which illustrates a high flammability range and low required energy for ignition. The difference in lower heating value (LHV) of H₂ has a direct impact on the economics of substituting NG with H₂. While the gravimetric LHV of H₂ is higher than that of CH₄, the relation turns when volumetric LHVs are compared. Thus, under the same operating conditions, the amount of energy delivered by equal volumes of H₂ and CH₄ is only around a third in the case of H₂ compared to CH₄.

2.2. Characteristics of the existing natural gas infrastructure

The understanding of the structure and boundaries of the system is crucial when investigating its repurposing for H₂ transportation. In the subsequent case study, we focus on the NEL pipeline in Germany. Hence, we highlight some characteristics of German and European natural gas pipeline infrastructure in this section.

The primary purpose of so-called transportation pipelines is the long-distance transport of natural gas. This is achieved in high-pressure pipelines, typically of large diameters. They are clearly distinguished from low-pressure, small-diameter pipelines in the distribution network, which serve (smaller) final customers.

The secondary task of the gas grid is to function as a short-term storage, providing flexibility to cover peak demands, the so-called “linepack flexibility”. It is achieved with the large amount of gas which is in the pipelines at any moment in time [17]. By adjusting pressure levels the amount of gas stored in the transportation system can be varied. Linepack flexibility is particularly high for transportation systems due to high operating pressure and pipeline volumes [18].

If the maximum allowable operating pressure (MAOP) exceeds 16 bar, pipelines are considered as “high-pressure”.⁴ Large transportation pipeline networks are typically operated under pressure up to 100 bar to realize large delivery capacities while they reach diameters up to 1400 mm [19]. Materials used are predominantly low-carbon and low-alloy steels in which up to 99 wt% is iron [20]. According to the European Gas Incident Data Group (EGIG), four material grades are the most installed: grade B, X52, X60 and X70 [21]. These grades are assigned by the American Petroleum Institute (API), where grade B is below grade X, and the following number represents the yield strength in pounds per square inch [21]. The movement of gas is achieved through pressure differentials in the pipelines [22]. Gas moves from higher pressure towards lower pressure and, thus, has to be recompressed periodically [19, 23].

The velocity of gas flow in pipelines is not limited in theory.⁵ However, an excessive velocity causes safety concerns as collisions of gas molecules with pipeline walls increases the internal erosion and the chances of material failure. Following Abbas et al. [23], the “erosional velocity” is reached at 20 m/s for natural gas. Mischner and Fasold [24]

³ The exact gas composition of the well stream varies depending on the exploitation site but is in the range of 0.75–0.99 M fraction [41].

⁴ See the German high-pressure gas pipeline regulation GasHdRLtGv, §1 (https://www.gesetze-im-internet.de/gashdrltgv_2011/BJNR092800011.html), accessed on April 12, 2024.

⁵ [23] argue that “the maximum speed any gas can travel is limited by the speed at which sound can travel in the same gas”.

state that a lower velocity of under 10–15 m/s is the common limit for natural gas flow in pipelines in Germany. However, Mischner and Fasold [24] also indicate that the erosional velocity is gas- and pipeline dependent, thus velocities up to around 30 m/s may be safe within some dust-free pipes.

2.3. Challenges of H₂ transportation in natural gas pipelines

In this section, we briefly discuss the challenges associated with hydrogen transportation in pipelines at large scale. We focus on two main aspects: first, the question whether pipeline transportation will be able to deliver enough H₂ to supply the large volumes that are potentially needed to replace natural gas consumption. Second, we discuss whether the transportation in pipelines can be done safely.

2.3.1. Ensuring security of supply

The lower energy content of H₂ compared to CH₄ (Table 1) might pose some challenges regarding the energy delivery capacity of repurposed pipelines. The argument may be raised that repurposed infrastructure will not be able to satisfy future demand for H₂. Galyas et al. [25] calculate that the loss in transmissible energy of pure H₂ amounts to around 18% at 25 bar and 23% at 75 bar compared to NG. Quarton and Samsatli [18] conclude with even less favorable values, namely a 36% energy loss at 80 bar inlet pressure in a study investigating a potential switch from NG to H₂ in the United Kingdom. Galyas et al. [25] also point out two other relevant aspects when switching from NG to H₂, namely the increase in flow velocity of the gas by 165% and an increase of necessary compression power by 260% (while maintaining the same operating conditions). The latter is the result of the less favorable compressibility characteristics of H₂ and the higher resulting volumetric flow rate compared to NG.

Moreover, linepack flexibility is severely impacted by the lower energy content of H₂. At an inlet pressure of 80 bar, the amount of stored energy in the pipeline network decreases by over 80% compared to pipelines operated with NG [18].

2.3.2. Ensuring safe pipeline operation

Operating pipelines safely while transporting pure H₂ is the second main area of concern when considering pipeline repurposing [26]. Hydrogen induced steel embrittlement (so-called hydrogen embrittlement) is labelled “a ‘chief technical concern’ in the pipeline transportation of hydrogen gas” [23]. This phenomenon has a particularly strong impact on pipeline welds [8]. Van der Zwaan et al. [27] describe the process as H₂ molecules dissociating at the surface of the steel into two hydrogen atoms (H) and migrating deep into the material. Once penetrated, the H atoms can recombine in micro voids to molecular H₂ gas and build up pressure inside the material, eventually causing failure. The extent of material degradation due to hydrogen embrittlement is dependent on multiple factors. The composition of steel plays a significant role, where types with high carbon and manganese content are more severely affected, as well as intensity and frequency of pressure swings during operation [27,28]. In other words, not all pipelines might be suitable for H₂ transportation, depending on the material.

Moreover, the wide range for flammability of H₂ in air and its high diffusion coefficient make handling of the gas challenging. In comparison to NG, H₂ escapes more easily from valves, seals, gaskets and fittings, but also through solid materials like pipeline steel [28]. The losses are likely small in economic terms, i.e., in relation to value lost. However, the real problem of insufficient containment is the risk of explosion. Leak detection is also part of the challenge, as hydrogen is a colorless and odorless gas. Addition of odorants is difficult as they need to have the same dispersion rate as H₂ to be able to escape with it [29]. Hence, odorants of natural gas cannot be used. In addition, once ignited, H₂ burns with a hardly visible flame at daylight [28].

Summing up, several challenges come with the use of natural gas pipelines for hydrogen transport. Thus, it must be investigated which

elements of natural gas pipelines need to be repurposed and with which technology at what cost.

3. Design of a repurposed H₂ pipeline infrastructure

In this section, we investigate which criteria must be met by (former NG) pipelines if they are used for H₂ transportation. We start by investigating which elements of the pipeline system need particular attention before examining how the amount of delivered energy of H₂ can be ensured to be sufficiently high. Lastly, we introduce the three pipeline repurposing options that are relevant and feasible and that we apply to the NEL case study in Section 4.

3.1. Elements being subject to change during repurposing

NG pipelines can potentially be used for H₂ transport without a quality loss in terms of H₂ purity [30]. However, several key elements of the NG transportation system must be adapted if they are to be used to transport H₂. The components impacted most severely are metering devices, valves, and compressor stations. The high-pressure steel pipes themselves also need to be adjusted to a certain extent [31] as we discuss below (Section 3.3).

The gas distribution system is likely to be less impacted by a switch to H₂. This is due to the compatibility of commonly used plastic pipelines (PE or PVC) and lower operating pressure [32]. The elements to be replaced in the distribution grid are mostly end-user devices in households, industry, and power generation. Hence, in the following, we focus on transportation pipelines.

3.1.1. Compressor stations

Pressure differences in pipelines are responsible for the movement of gas, which flows from higher to lower pressure. To achieve a pressure differential between two points, large compressor stations are located along the system to periodically recompress the transported gas. Two compressor types are commonly used, namely centrifugal and piston (reciprocal) compressors, usually composed of multiple units in series or in parallel [17,19]. The compressor type used depends largely on the operating conditions. Reciprocal compressors are considered a less efficient solution for handling very large amounts of gas [6]. However, reciprocal compressors are less sensitive to gas density and, thus, are likely better suited for handling hydrogen [33,34].

Existing compressor stations are not necessarily suited to operate with 100% hydrogen for multiple reasons. First, existing compressor stations are often fueled with natural gas, self-consuming some of the transported gas in the pipeline [33]. If no more natural gas is transported in the pipelines, the fuel supply becomes an issue and compressor stations would need to be fueled by hydrogen, electricity, or other fuels.

Second, the lower density of hydrogen and, consequently, the higher volumetric flow rate potentially lead to failure of the impellers in centrifugal compressors [32,33]. Although this issue only concerns centrifugal compressors, it still cannot be neglected as they represent most installed compressors in gas transportation infrastructure. This challenge can be overcome and some manufacturers started producing dedicated radial type turbo compressors suited for large H₂ volumes [32, 34–36].

3.1.2. Leakage prevention and metering devices

The risk of hydrogen leakage is more significant in places where different components are joined together. The largest units concerned by this issue – besides compressor stations – are the valves (in large pipelines predominantly ball valves), installed every 32–48 km along the pipelines [37]. Valves enable maintenance works and the identification of leaks by shutting off separate pipeline sections or releasing pressure if required. In case of an emergency, the valves are used to completely shut off the affected sections [38]. Generally, valves installed in gas pipelines face the same challenges as pipeline materials. Due to the physical

properties of hydrogen – particularly the very small molecular size – valves may not be able to completely seal off pipeline sections and, thus, not fulfil their purpose anymore. Bard et al. [31], therefore, indicate that valves require significant changes during the repurposing process.

Metering stations are installed in pipeline networks at entry and exit points. The purpose of metering stations is to provide transparency for grid operators about the amount of gas present in their pipelines and, thereby, enabling correct management of the system [37]. Currently, gas grids utilize the principle of volumetric flow management. Metering devices are calibrated according to the energy content of natural gas, thus giving an appropriate indication of the energy amount delivered and/or stored in a pipeline section [39]. These devices need to be recalibrated due to the differing properties of hydrogen.

3.2. Solutions to ensure security of supply

Solutions to mitigate the possible issues resulting from a lower amount of transported energy in repurposed gas pipelines aim at increasing the pipeline network's capacity. One option could be to increase the operating pressure of existing pipelines and thus increase the volumetric energy density of the transported hydrogen. Centrifugal and reciprocal compressors both are able to generate far higher pressure levels than the commonly used pressure levels in gas pipelines [40]. Other options are to install additional compressor stations or to use looping in specific sections [41]. These measures are either CAPEX-intensive or carry additional safety risks due to higher operating pressure. However, the necessity of such changes remains debatable. For the case of Germany, for example, the future demand for H₂ is expected to be far below the capacity of the current gas transportation infrastructure in many studies due to the widespread electrification of final energy use (e.g. [42,43]).

3.3. Options for pipeline repurposing

The most critical task during repurposing is to ensure safe operations of the pipes. As of now, multiple repurposing options are known and discussed in industry, while none has yet been rolled out. In the following, we discuss the four major options. They differ significantly and thus require unique modifications of the existing infrastructure. Hence, the suitability of options may be different for individual use-cases.

3.3.1. Enhanced monitoring and maintenance

The most trivial approach to repurpose pipelines is keeping the existing pipes unchanged. Then, monitoring and maintenance measures need to be enhanced (i.e., increased in frequency) to detect cracks, leaks, and other potential dangers as early as possible [12]. This approach seems reasonable as hydrogen embrittlement is a slow process and the failure of pipeline materials would not occur immediately, but only after a certain amount of time under operation. This option also appears feasible, as multiple materials deemed suitable for the transportation of hydrogen are also listed in the standards for pipeline materials for high pressure natural gas transportation pipelines (DIN EN ISO 3183; [32]). Yet, the faster degradation of steel in contact with hydrogen is unanimously acknowledged. Fatigue crack growth rate in pipelines transporting hydrogen is 5–15 times larger than in pipelines which transport natural gas [12]. To account for this, we consider the reduction of expected pipeline lifetime by 25% as proposed by Neugebauer [32]. Neugebauer [32] also raises the point that pipeline integrity management systems (PIMS) for gas transportation networks must be adapted, especially to ensure the integrity of welds and fittings. This leads to higher costs of monitoring and maintenance of H₂ pipelines compared to NG pipelines.

3.3.2. Admixing degradation inhibitors

The purity of the H₂ stream or, in turn, the admixture of other gases

can influence the extent of hydrogen embrittlement in pipeline steels [44]. Some gases like oxygen (O₂), sulphur dioxide (SO₂) and carbon monoxide (CO) inhibit H₂ embrittlement [45,46]. The inhibition is achieved through admixed gases blocking the steel surface and preventing the absorption of hydrogen into pipeline material. The required concentration of inhibitor gas depends on the admixed gas and can be low for some gases. While SO₂ and CO limit H₂ embrittlement at a concentration level of 2%, the same effects could already be observed at a lower concentration level of 0.015% of O₂ [12]. We use oxygen in the following sections as representative inhibitor gas which has a low admixture share requirement and is favorably assessed in the literature.

There are several limitations to admixing gases to hydrogen which need to be considered. Most importantly, the mixtures need to be combustible, non-toxic when burned, compatible with end user devices, and non-explosive. The use of O₂ raises concerns regarding the explosivity of the gas mixture under certain circumstances. To avoid this problem, the percentage of oxygen needs to stay outside of the flammability limits of H₂ in pure oxygen, which is between 4.65 and 93.9 vol % H₂ [47]. The required 0.015% of O₂ is clearly below the flammability limit, thus, we do not consider explosion hazard as an exclusion criterion for its use in H₂ pipelines.

Certain end use applications require extremely high purity hydrogen, which might be compromised by mixing inhibitor gases into the transported gas. One prominent example of such a H₂ use are fuel cells in H₂-driven vehicles [29,48]. This potentially requires purification of hydrogen before being delivered to consumers with, e.g., a pressure swing adsorption process [12]. However, there are two significant arguments against the use of centralized purification units at the exit point of the pipeline. Firstly, the purification process is very energy intensive and costly [49]. To reduce the resulting costs, only as much hydrogen as necessary should be purified. Several H₂ use applications, such as gas turbines, are far less sensitive to impurity, thus not requiring or justifying centralized purification. Second, some consumers that will require very pure hydrogen may already do so today. Thus, it is reasonable to assume that they already have their own small-scale purification facilities in place. So, centralized purification units would not be required.

From a cost perspective, there are no apparent reasons which would indicate the necessity of excavating existing pipelines when admixing inhibitor gases. This reduces fixed (capital) costs of this repurposing approach significantly, together with low intensity of labor. However, there is no concluding evidence yet whether full inhibition of H₂ embrittlement can be reached by admixing gases and further research is necessary [50]. In addition, adding gaseous inhibitors would require coordination between individual line segments.

3.3.3. Pipe-in-pipe

During the pipe-in-pipe (PIP) approach, an inner pipe is inserted into the existing steel pipeline to form a physical barrier between steel and transported gas [51]. The inner pipe is made from a material capable of withstanding traditional difficulties posed by hydrogen, while the strong outer pipeline is already designed to withstand the pressure of the transported gas. Thus, the inner pipeline is not required to be a structural component, leading to cost savings in comparison to newly built infrastructure.

Several technologies are known for the PIP approach with lined pipes being the most promising. In this case, metallic or non-metallic protective materials are fitted into the tubular section of the pipelines either at the end of the pipeline manufacturing process or as a rehabilitation measure [52,53]. Especially the latter appears interesting for pipeline repurposing projects.

Most used liners in industry and in oil and gas installations are made from polyethylene and are divided into three classes depending on requirements they must fulfil [52]. Liners used for so-called structural rehabilitation are designed to completely withstand the pressure inside the pipeline, while the ones used for semi structural rehabilitation can only withstand the pressure partly. Both options are designed to resolve

Table 2
Technical operating parameters of the NEL pipeline.

Parameter	Value	Source
Pipeline length	441.46 km	(1)
Technical availability capacity p.a.	20 bcm	(1)
Inlet pressure ^a	100 bar	(3)
Outlet pressure	79.88 bar	(4)
Pipeline outer diameter	1422 mm	(2)
Pipeline inner diameter	1376 mm	(2)
Pipe material	X70 steel	(3)
Number of compressor stations	0	(1)

^a Assumed to be equal to the maximum allowed operating pressure of the pipeline.

^b Fluxys, NEL facts and figures: <https://www.fluxys.com/en/company/fluxys-deutschland/nel-eugal-> accessed March 20, 2023.

^c Europipe: <https://www.europipe.com/de/referenzen/referenzprojekte#1538> – accessed March 20, 2023.

Sources: (1): Fluxys^b, (2): Europipe^c, (3) [12]; and (4): own calculations

Table 3
Results for pipeline capacities.

Parameter	Natural gas	Hydrogen
Energy delivered p.a.	185.75 TWh	137.86 TWh
Mass delivered p.a.	13.36 Mt	4.14 Mt
Gas volume delivered p.a.	20 bcm	49.43 bcm
Erosional velocity	11.49 m/s	37.06 m/s
Maximum actual gas velocity in the pipeline	4.81 m/s	17.76 m/s

structural weaknesses of the pipe in question and appear not to be especially relevant for pipeline repurposing, as the pipes to be repurposed are expected to be intact. Rather, non-structural liners are most suited, which solely serve as a physical barrier between the pipe and the gas as discussed above. Despite the good durability of polyethylene in theory, long-term performance data (for over 20 years of operation) is lacking, which leads to some uncertainty when considering this approach for repurposing gas pipelines for hydrogen transportation. From an economic perspective, the main cost driver of the PIP approach is material costs for the inserted liner, as there are no indications that the pipelines need to be excavated during the insertion process.

3.3.4. Inner coating of steel pipeline

Inner coating of gas transportation pipelines aims at inhibiting the harmful effects of hydrogen on pipeline materials similar to the pipe-in-pipe approach. An internal barrier is created by applying a non-corrosive material on the inner pipeline wall.

One of the commonly used technologies is so-called fusion-bonded epoxy (FBE) coating which first saw its commercial application in the late 1950's. FBE is applied in production plants via “the electrostatic application of micron sized thermosetting powders onto heated steels” [54]. During the process, the powders melt and form a protective layer on the steel surface until it finally cures. The thickness of the protective layer is usually around 400 µm to achieve an optimal coating performance [55]. Aluminum or zinc are also suitable for the inner coating of pipelines as they offer resistance against hydrogen embrittlement depending on the alloy [51]. A coating layer of aluminum oxide reduces the permeability of a steel pipeline by the factor of 10,000 and would only have to be 1–2 µm thick, thus, also being attractive from an economic point of view [51].

While it seems advantageous to use a coating layer to prevent the degradation of pipeline steels, the feasibility in existing pipelines is unclear [51]. Currently, the coating layer is applied during the production process of the pipes. It is uncertain whether the application of the layer is possible once the pipelines are installed and, hence, whether this process is an option for repurposing. This applies to FBE coating equally as for aluminum. Therefore, we do not consider this approach in

the remainder of this paper, but the technology might become relevant in the future.

4. Cost comparison of repurposing options in a case study

In this section, we apply the economic aspects of pipeline repurposing and calculate and compare the levelized cost of transportation (LCOT) of the available options in a case study investigating the potential repurposing of the “Nordeuropäische Erdgasleitung” (NEL).⁶ The North European natural gas pipeline (NEL) originally was connected to the Nordstream pipeline in Lubmin, Germany, together with the EUGAL and OPAL, two other major connecting pipelines of Nordstream in Germany. NEL is connected to the MIDAL pipeline after crossing Germany in East-West direction. NEL is the only pipeline in northern Germany in East-West direction.⁷ When NEL was opened, it significantly reduced the transport distance of Russian gas to the Netherlands and Belgium. Although it appears unlikely that the NEL will be repurposed anytime soon, it is an interesting candidate for the analysis of future repurposing projects due to its material properties which are relatively well documented compared to most other pipelines and due to its simple setup with only one entry and one exit point. It should also be noted, that especially in early phases of the development of a H₂ transportation network, significantly lower pressures than used in current gas pipelines could be sufficient to satisfy demand. However, our analysis is for the hypothetical case that repurposed pipelines will be required to operate at similar pressures as the current gas infrastructure, once the H₂ market has ramped up in the future.

4.1. Technological aspects

NEL was commissioned in November 2013. Due to its recent construction date, the pipeline can be considered state-of-the-art technology and material. The pipeline only has one entry and one exit point, each on one end of the pipeline. The absence of further entry- and exit points allows for a comparatively simple calculation of the necessary pipeline parameters. This applies for the pipeline's initial state as well as for calculating the repurposed pipeline's characteristics, including costs of repurposing. As not all necessary information to analyze repurposing of the pipeline is publicly available, a set of calculations based on Menon [56] is performed to obtain missing values. For reasons of simplicity, an isothermal flow in the pipeline and an average flow temperature of 10 °C are assumed. This is reasonable as gas can be expected to settle at ambient temperature in a long pipeline like the NEL.

The parameters for the NEL operated with natural gas are listed in Table 2. The downstream (outlet) pressure is calculated by solving the general flow equation as proposed by Ref. [56] and given publicly available data for the pipeline. The erosional velocity is calculated at 11.49 m/s and the gas flow velocity reaches a maximum of 4.8 m/s which are below the discussed safety margin (cf. Section 2.3).

We assume that the necessary changes during the repurposing process are intended to be kept as low as possible to minimize emerging costs. Thus, the pipeline characteristics such as inlet and outlet pressure are kept at the same values as they were during the operation with natural gas. The equations presented by Ref. [56] used for the calculation of the values listed in Table 2 can be adapted for the repurposed pipeline if the gas specific values and constants are changed accordingly. Hence, we use the given characteristics. In addition, we assume a limit for the gas velocity of 50% of the erosional velocity for the NEL operated

⁶ NEL Gastransport GmbH: <https://www.nel-gastransport.de/>.

⁷ Since the end of gas imports via Nordstream in 2022, there have been plans to reverse the flow on the NEL pipeline. See, e.g., the German network development plan 2022–2032 (FNB Gas (2024): [Netzentwicklungsplan Gas 2022–2032](https://www.netzentwicklungsplan-gas.de/). 2024.03.20_NEP-2022_Gas_FINAL_DE.pdf (fnb-gas.de), accessed on May 13, 2024).

Table 4

Compressor dimensions for H₂ compression.

Parameter	Value
Required compression power	232.75 MW
Energy required by compressor ^a	245 MW
Annual power consumption	1.97 TWh
H ₂ self-consumption ^b	2.14%

^a Including mechanical efficiency losses of the compressor.

^b Percentage of annual transported hydrogen and a gas turbine compatible with hydrogen with an efficiency of 50% to power the compressor.

Table 5

Economic input parameters for the LCOT calculations.

Parameter	Value	Source
Pipeline CAPEX	1 b€	(1)
Pipeline OPEX	1% of CAPEX p.a.	(2)
Pipeline OPEX (enhanced maintenance option)	5% of CAPEX p.a.	(2)
Compressor CAPEX (10 MWel)	1,228,000 €/MWel ^a	(3)
Compressor CAPEX (5 MWel)	1,613,000 €/MWel ^a	(3)
Compressor CAPEX (1 MWel)	3,017,000 €/MWel ^a	(3)
Compressor OPEX	4% of CAPEX p.a.	(3)
Pipe in pipe CAPEX	648.5 €/m liner ^a	(4)
Inhibitor OPEX	0.06 €/kgO ₂	(2)
Pipeline economic lifetime	40 years	(2)
Pipeline economic lifetime (reduced)	30 years	(5)
Compressor lifetime	20 years	(2)
Electricity price	100 €/MWh	(5)
WACC	6%	(6)
Inflation	2%	(5)

^a Adjusted with 2% inflation p.a. to 2023.

^b <https://www.europipe.com/de/referenzen/referenzprojekte#c1538>,

accessed March 20, 2023.

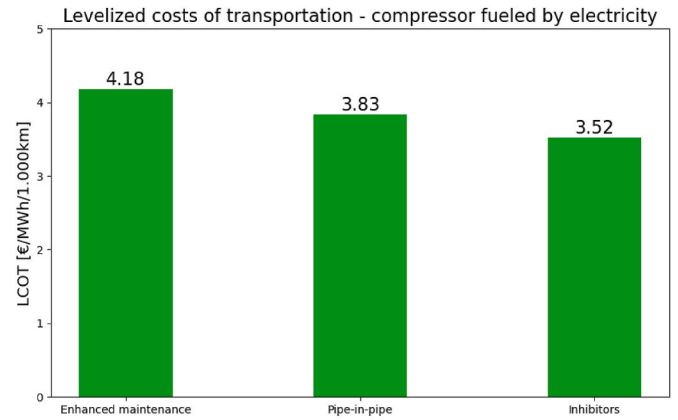
^c The cost assumption for the consumed hydrogen is based on the “Global Hydrogen Review” published by IEA where production costs of renewable H₂ are estimated to range between around 2 and 8 USD/kgH₂ in 2030 in the “Net Zero Emissions by 2050” scenario [60].

Sources (1): Europipe^b, (2): [12], (3): DNV GL^c, (4): [51], (5): own assumption, (6): [59].

with H₂ which we calculated. This constraint, however, does not appear to be binding, as the erosional velocity is not reached for the given pressure differential.

Table 3 compares the calculated capacities of the NEL pipeline operated with natural gas or hydrogen. As expected, the energy delivery capacity of the repurposed pipeline decreases with hydrogen compared to the transport of natural gas. The higher gas velocity and subsequent larger gas volume are not enough to offset the lower volumetric energy density of H₂.

The NEL is currently not equipped with a compressor station as the natural gas used to arrive with sufficiently high pressure from Nordstream. However, we cannot assume that this will be the case for hydrogen after repurposing the pipeline. Especially if electrolytic hydrogen was to be transported through the repurposed NEL, gas compression would be necessary at some point. Current commercial PEM electrolysis produces hydrogen at a pressure of 30–40 bar [57]. As there is ongoing research for high-pressure PEM systems, we use the higher end of this range and assume that a compression from 40 bar to the NEL inlet pressure of 100 bar will be necessary. To calculate the dimensions of the required compressor in Table 4, we use the general flow equation and mass balance combined with the ideal gas equation as presented by Ref. [56] in combination with the values in Table 3. The Friction factor within the general flow equation is calculated using the Colebrook-White equation. It should be noted that the pressure differential between the electrolysis output (40 bar) and the pipeline inlet pressure (100 bar) is unconventionally large. Hence, compression is very energy intensive. We approximate the adiabatic efficiency of the



LCOT by cost component - compressor fueled by electricity

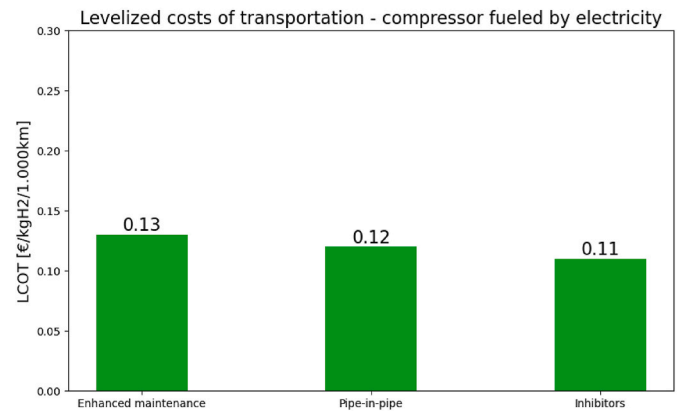
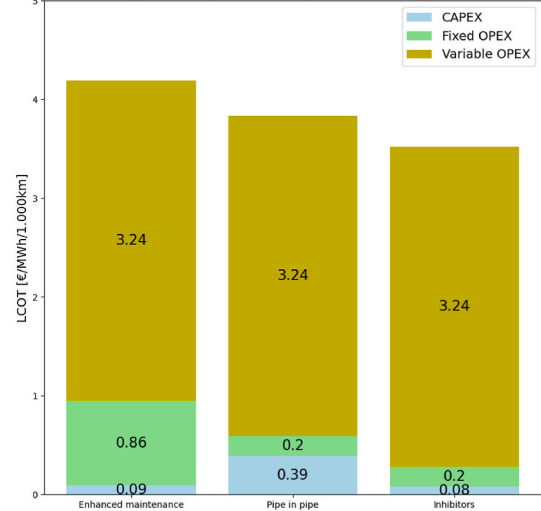


Fig. 1. LCOT repurposed pipeline (compressor fueled by electricity).

compressor at 80% which is in the range of 75%–85% reported in the literature [56]. Mechanical compressor efficiency is assumed to be 95% which is also a commonly used value in literature. It shall be noted that in the case of a real-world application additional cooling units would be included in the compressor station, which are not accounted for in our calculation.

One large challenge for future hydrogen infrastructure may be the energy required by compressor stations which are often fueled with natural gas today. Their operation with electricity represents the most efficient solution. However, making the required power available may pose significant challenges for power grid operators. A compressor station is a large and inflexible demand which would also accrue during times of low electricity availability, as these are the times when demand

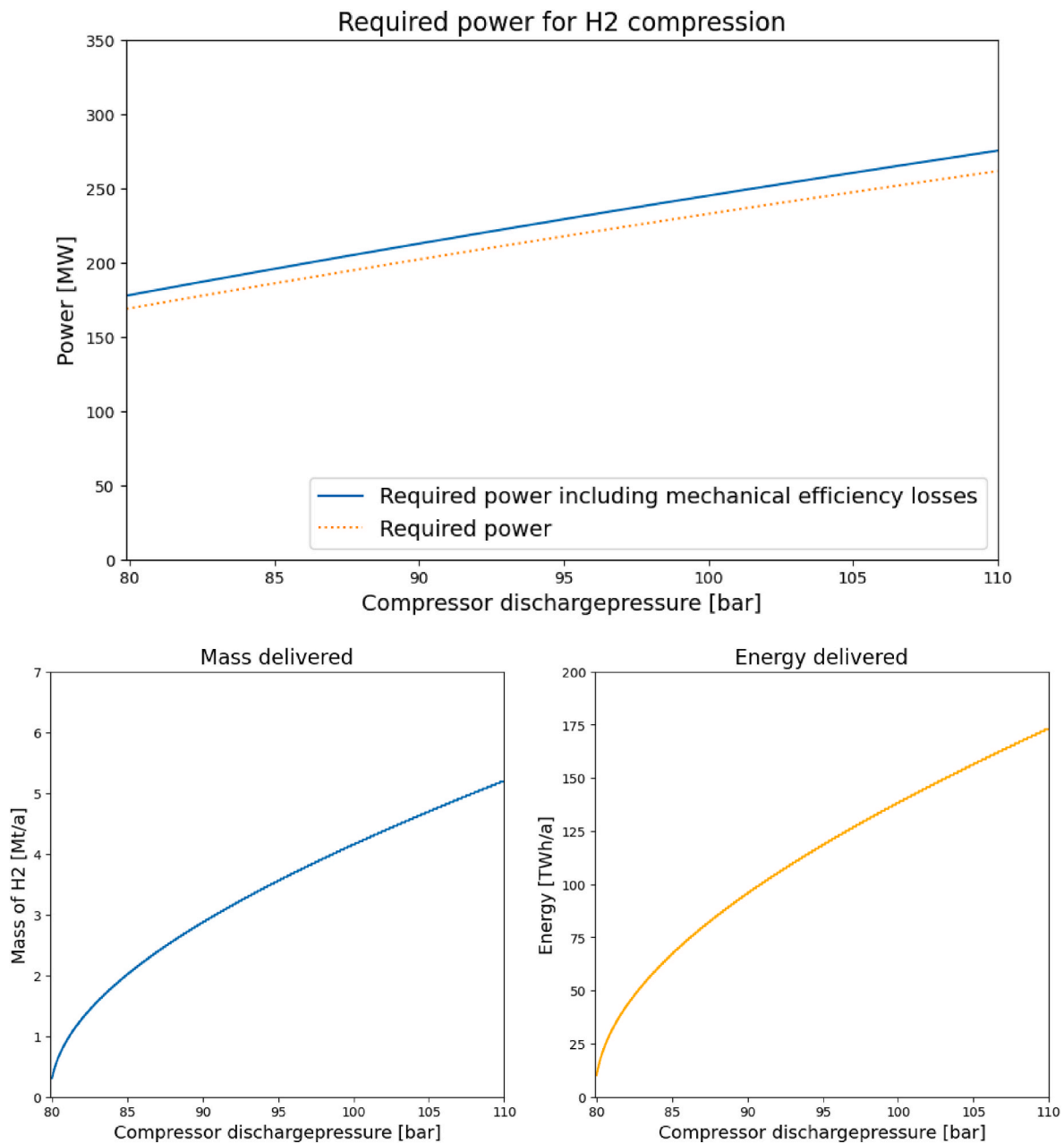


Fig. 2. Compressor layout depending on pipeline inlet pressure (top) and resulting pipeline capacity (bottom).

for hydrogen rises. Another option is to power the compressors with some of the transported gas itself, as is currently the case in NG infrastructures. We calculate a self-consumption level of 2.14% based on the energy needs of the compressor and the heating value of the H_2 . This would considerably exceed the self-consumption of current gas compressors. For reference, a compressor for a pipeline with identical operating parameters which transports natural gas, would annually consume 0.54% of the transported gas.⁸

The technological pipeline and compressor simulations are carried out using Python.⁹ We validated our calculations with the commercial simulation software Aspen HYSYS that is used by the gas industry.

During this procedure, a two-stage piston compressor system with intermediate cooling was implemented in the software.

4.2. Economic aspects

We compare the three repurposing options based on their levelized transportation costs (LCOT). We adapt the definition of levelized costs of electricity (LCOE) presented by Ref. [58]. We calculate the LCOT for the length of the NEL, 441.46 km, and normalize the results to obtain values comparable with other public studies in the unit of €/MWh/1000 km with the following approach:

$$LCOT = \frac{I_0 + \sum_{t=1}^n \frac{A_t}{(1+i)^t}}{\sum_{t=1}^n \frac{M_{t,H_2}}{(1+i)^t}} \times \frac{1000 \text{ km}}{441.46}$$

I_0 is the Investment expenditure in €; A_t is the annual total cost in € per

⁸ Identical operating parameters: Compression levels between 40 and 100 bar, gas turbine with an efficiency of 50% to power the compressor, pipeline capacity as listed in Table 3 with natural gas.

⁹ Python code available at <https://zenodo.org/records/11178736>.

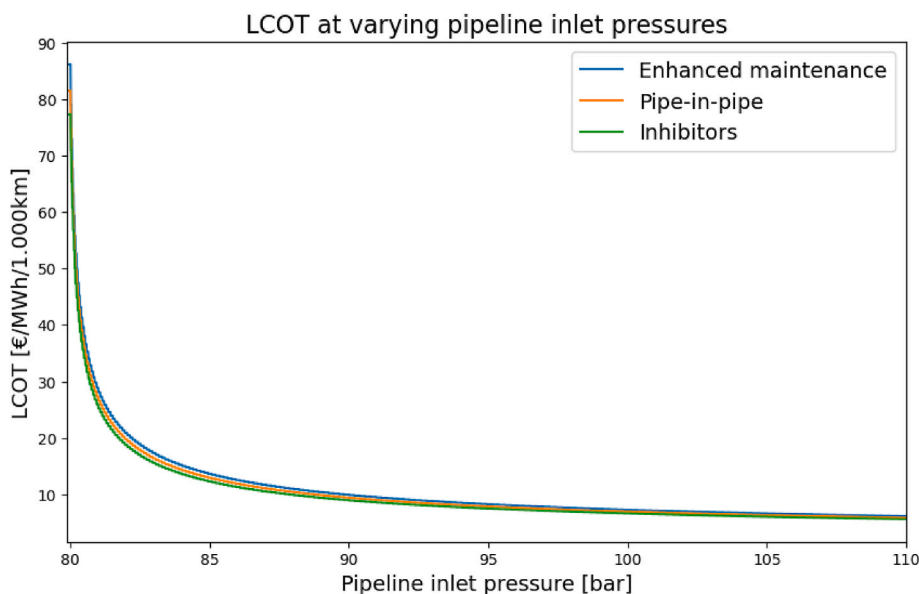


Fig. 3. LCOT at varying pipeline inlet pressure levels and with a compressor operated with electricity.

year t ; and M_t is the delivered amount of Energy in form of hydrogen through the pipeline in MWh per year t ; t is the year of lifetime (1, 2, ..., n) and i is the real interest rate in %.

We do not include the pipeline CAPEX in our calculation, as we investigate a pipeline which is already in place. However, we account for the OPEX of the pipeline, representing necessary maintenance measures. The values in the literature range between 0.8% and 5% of the CAPEX per year [12]. We use 1% if inhibitors or pipe-in-pipe are used for repurposing. This is slightly higher than the 0.8% currently applied for NG infrastructure in the German regulation¹⁰, and is chosen by us to account for uncertainties regarding the maintenance of future hydrogen infrastructure. For the case of repurposing through enhanced maintenance,¹¹ we assume an OPEX of 5% of the pipeline CAPEX p.a. to reflect more frequent maintenance measures. An identical value was used by Cerniauskas et al. (2020) when investigating the same option. In addition to higher operating expenses, we assume pipeline lifetime reduces by 25% for the enhanced maintenance option (cf. Section 1.1.4). As the compressor station is a greenfield investment, we include CAPEX and OPEX in our calculations. The cost assumptions are retrieved from a database published by DNV GL in 2020.¹² The OPEX consists of fixed OPEX in the form of maintenance costs to which we add variable OPEX (i.e., fuel costs). Two options for pipeline repurposing bring individual cost elements in addition to the already discussed expenses. For the pipe-in-pipe approach, we account for the CAPEX of the inserted liner material and in the inhibitors approach, we include the costs for the inhibitor gas. The full list of the economic input parameters included in the LCOT calculations of the three repurposing options is given in Table 5.

To calculate the LCOT, the assumed pipeline lifetimes for the individual repurposing options are used as a time horizon. Assumptions regarding the extent of potential reinvestments after this time would carry high uncertainties, as there is currently no experience with the

integrity of repurposed pipelines after a long time of operation. The pipeline lifetime exceeds in every case the economic lifetime of compressor stations so that a major reinvestment measure is assumed after 20 years. DNV GL states that an additional 100% capital costs should be accounted for the greenfield installation of the compressor. Thus, at t_0 we calculate with 200% of the CAPEX listed in Table 5. At t_{20} , when the reinvestment takes place, we calculate with 100% CAPEX again.

The LCOT of the individual repurposing options are shown in Fig. 1. The cost difference between options is rather small. Inhibitors are the least expensive choice and enhanced maintenance is the costliest option. This is due to relatively low costs for the inhibitor gas (and because we do not assume a centralized purification unit). The shorter pipeline lifetime with the enhanced maintenance option leads to a significantly lower amount of gas transported over the total pipeline lifetime. This drives the costs per unit above the other options. On the righthand side of Fig. 1, the LCOT is split into cost components. We can observe that the variable OPEX, i.e. the fuel costs of the compressor, are by far the largest part of the expenses. This means that power prices have a significant influence on transportation costs. For example, with an electricity price of 50€/MWh, an LCOT of 1.90 €/MWh/1,000 km can be reached with inhibitors. At the bottom of Fig. 1, the LCOT are shown in € per kgH₂ transported.

As mentioned above, the variable compressor OPEX accounts for the largest part of the LCOT. Thus, adjusting the compressor layout might be a possibility to reduce costs. We investigate two alternative possibilities: 1) to reduce the pipeline inlet pressure, and 2) to change the fuel from electricity to self-consuming parts of the transported hydrogen. Concerning option 1, we still assume a setup where the H₂ is supplied at 40 bar (e.g. from PEM electrolysis) to the compressor at the pipeline entry point and keep the pipeline outlet pressure the same as before repurposing (79,88 bar, cf. Table 2) to minimize the necessary changes to the pipeline. If this outlet pressure is to be reached the pipeline inlet pressure needs to be in a range between 80 and 110 bar. We acknowledge that 110 bar is above the current maximum allowable operating pressure (MAOP) of the NEL and thus is only used to illustrate the effects of increasing pressure in the pipeline.

As expected, there is a positive correlation between decreasing the pipeline inlet pressure (which is synonymous to the compressor discharge pressure) and the amount of delivered H₂ for a constant pipeline outlet pressure. At the same time, the energy requirement of the compressor also decreases. The compressor sizing to reach varying

¹⁰ <https://www.europipe.com/de/referenzen/referenzprojekte#c1538>, accessed March 20, 2023.

¹¹ The cost assumption for the consumed hydrogen is based on the “Global Hydrogen Review” published by IEA where production costs of renewable H₂ are estimated to range between around 2 and 8 USD/kgH₂ in 2030 in the “Net Zero Emissions by 2050” scenario [60].

¹² Incentive Regulation Ordinance (ARegV), §23.https://www.gesetze-im-internet.de/aregv/_23.html, accessed February 01, 2024.

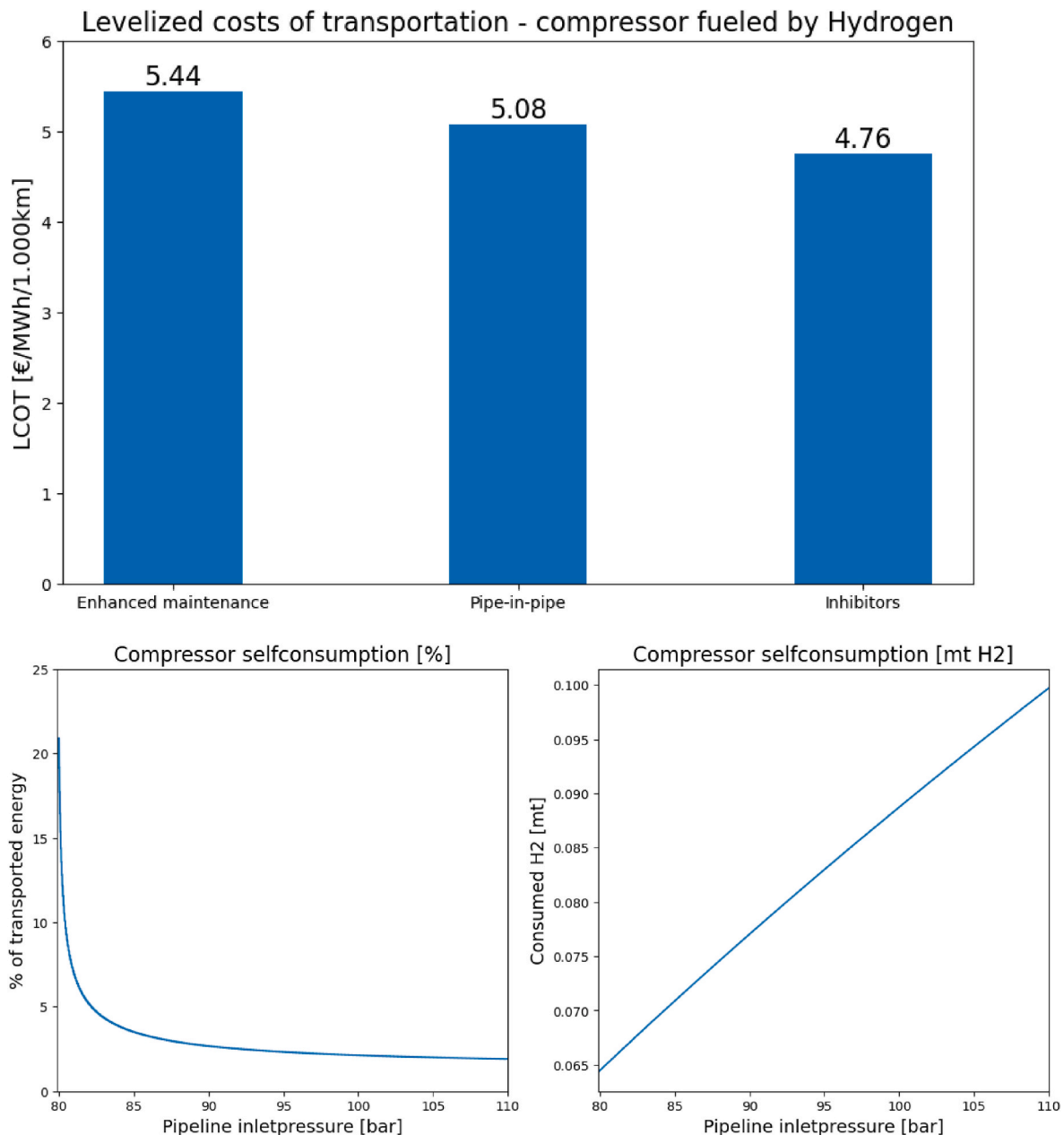


Fig. 4. LCOT - compressor fueled with H₂ (top) and amount of H₂ consumed depending on pipeline inlet pressure (bottom).

discharge pressure levels ranges from 177.65 MW to 275.28 MW. Depending on the compressor layout, the transported energy (mass of H₂) ranges from 10.28 TWh/a (0.31 Mt/a) to 172.84 TWh/a (5.19 Mt/a). As can be seen in Fig. 2, the curves of delivered energy and mass are steeper in the lower pressure regions and become flatter when approaching the upper end of the investigated pressure range.

Compressor size and pipeline inlet pressure influence the LCOT in certain ways. However, the benefit of delivering more H₂ due to a higher inlet pressure outweighs the additional costs caused by a larger compressor, if the pipeline geometry and outlet pressure are unchanged (see Fig. 3). LCOT fall with increasing pressure levels, but there is no change in the cheapest repurposing option, independent of the pipeline inlet pressure.

Switching the compressor fuel from electricity to hydrogen via self-consumption increases the LCOT for the initial configuration with 100 bar inlet pressure (see Fig. 4). We account for the H₂ used by the compressor in the form of lost transmissible energy and by attributing a

cost of 3€/kgH₂ to it.¹³ Assuming higher H₂ prices would lead to higher LCOT. Putting into relation the percentage of transported gas used by the compressor and the total amount of transported hydrogen leads to the same LCOT ranking as in the case of an electricity fueled compressor. Also, while the absolute value of consumed gas rises with increasing pipeline inlet pressure, the percentage of the transported gas being consumed decreases. It reaches as much as 20.85% at 80 bar inlet pressure, decreasing to 1.92% at 110 bar. This leads to the conclusion that from an efficiency point of view, the pipeline operation at a high-pressure level is favorable.

¹³ We assume that a conversion of the NEL natural gas pipeline to a hydrogen pipeline without adaptation measures is feasible, based on studies like [61] who find for a similar steel grade (X65 in their case, compared to X70 in NEL) that “the impact of [...] hydrogen on this grade are moderate.”

5. Discussion and conclusions

This study assesses the main technical and economic aspects of repurposing existing high-pressure gas pipelines for hydrogen transportation with a case study of the NEL pipeline in Germany. Due to the characteristics of H₂ there are several challenges regarding the safe operation of repurposed pipelines (steel corrosion, low energy density, risk of diffusion and explosion, etc.) which must be solved. However, none of the identified problems seems to be impossible to overcome. The problems are well documented and the technologies to mitigate them already exist.

One of the most significant challenges which this study identifies is the newly emerging (power) demand that results from compressor stations along the repurposed pipelines. The compressor station in our case study would have roughly the same electricity demand as a small city, bringing new challenges for the electricity grid. The electricity infrastructure must be built in the first place, as such a large amount of electricity is not required with current natural gas compressors. This challenge needs to be addressed as soon as possible by electricity and gas grid operators jointly to enable the repurposing of gas pipelines for H₂ transport. Self-consumption of hydrogen may be an alternative, depending upon the valuation of the gas.

The levelized costs of transportation that we calculate in this study are around the same order of magnitude as in other public studies. An overview of studies collected by the European Union Agency for the Cooperation of Energy Regulators [6] finds LCOT between 2.4 and 6.17 €/MWh/1,000 km which includes the range of values calculated in this study (3.52–4.18 €/MWh/1,000 km; cf. Fig. 1). However, it is not fully transparent which approaches for pipeline repurposing were chosen in the studies cited by Ref. [6]. In contrast, we aim at providing a transparent cost calculation.

The potential technical pipeline capacity of the repurposed NEL of 138 TWh per year seems very large. It would be enough to satisfy over the half of the forecasted total gross hydrogen demand in Germany in 2045, which is expected to be 226 TWh according to a study by the German energy agency [42]. This leads to the conclusion that large repurposed natural gas pipelines, such as the NEL, will not need to be operated at their technical limits to satisfy future demand, especially in the early stages of adoption.

Out of all three repurposing options that we assess in our study, the use of gaseous inhibitors seems to be the most economic. However, the other two options are only slightly more expensive options. This leads us to conclude that aspects other than just monetary considerations will play a significant role when deciding on a repurposing option of a specific pipeline. One of these aspects may be the social acceptance and perceived risk of the option by citizens. It is possible that physical enhancement of pipelines with the pipe-in-pipe approach may raise a higher feeling of security amongst the general population than the use of gaseous inhibitors or enhanced maintenance. However, social acceptance and the perception of security for repurposing measures is a topic not well understood in the literature, thus leaving room for future research. At the same time, the high fixed costs of investments into individual repurposing options may be a cause of concern for some infrastructure operators. As the future of hydrogen and the market size are still subject to uncertainty, infrastructure operators could prefer to repurpose only with enhanced maintenance, thereby avoiding exposure from significant upfront investment. If the market turns out to be smaller than expected, less investment is sunk at the expense of higher operational cost once demand has realized. Further, we expect that technical requirements of H₂ consumers, e.g. on purity, may have a significant effect on the adoption of gaseous inhibitors for a particular pipeline repurposing project. Another interesting aspect is the choice of pipelines for repurposing. Pipelines with smaller diameters than the NEL could likely be operated at their current pressure levels while providing sufficient capacity to satisfy future demand. High pressure levels and differentials minimize the costs at fixed pipeline geometry and thus lead to

the most favorable LCOT. In all these areas, further detailed research is urgently needed.

CRedit authorship contribution statement

Kornél Télessy: Writing – original draft, Visualization, Software, Formal analysis, Data curation. **Lukas Barner:** Writing – review & editing, Validation, Supervision, Methodology, Conceptualization. **Franziska Holz:** Writing – review & editing, Validation, Supervision, Project administration, Methodology, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

The authors gratefully acknowledge funding from the Berlin-based Einstein Foundation for the project “Open-source modeling of the future role of renewable hydrogen in Germany and Europe” (grant no. A-2020-612).

References

- [1] Zachmann G, et al. Decarbonisation of energy: determining a robust mix of energy carriers for a carbon-neutral EU. DIW Berlin: DIW Berlin: Politikberatung kompakt; 2021. p. 175.
- [2] Schill W-P. Electricity storage and the renewable energy transition. *Joule* Oct. 2020;4(10):2059–64. <https://doi.org/10.1016/j.joule.2020.07.022>.
- [3] Nemmour A, Inayat A, Janajreh I, Ghenai C. Green hydrogen-based E-fuels (E-methane, E-methanol, E-ammonia) to support clean energy transition: a literature review. *Int J Hydrogen Energy* Apr. 2023;S0360319923013393. <https://doi.org/10.1016/j.ijhydene.2023.03.240>.
- [4] Jaro J, Wang A, van der Leun K, Peters D, Buseman M. Extending the European hydrogen Backbone. In: A European hydrogen infrastructure vision covering 21 countries. Guidehouse. Guidehouse; 2021. April 2021.
- [5] van Rossum R, Jens J, La Guardia G, Kühnen L, Overgaag M. European hydrogen Backbone. A European hydrogen infrastructure vision covering 28 countries. Guidehouse. Guidehouse; 2022. April 2022.
- [6] ACER. Transporting pure hydrogen by repurposing existing gas infrastructure: overview of existing studies and reflections on the conditions for repurposing. Eur. Union Agency Cooperation Energy Reg. 2021 [Online]. Available: https://acer.europa.eu/Official_documents/Acts_of_the_Agency/Publication/Transporting/20Pure/20Hydrogen/20by/20Repurposing/20Existing/20Gas/20Infrastructure_Overview/20of/20studies.pdf. [Accessed 3 February 2023].
- [7] Klopčič N, Stöhr T, Grimmer I, Sartory M, Trattner A. Refurbishment of natural gas pipelines towards 100% hydrogen—a thermodynamic-based analysis. *Energies* Dec. 2022;15(24):9370. <https://doi.org/10.3390/en15249370>.
- [8] Jia G, et al. Hydrogen embrittlement in hydrogen-blended natural gas transportation systems: a review. *Int J Hydrogen Energy* Sep. 2023;48(82):32137–57. <https://doi.org/10.1016/j.ijhydene.2023.04.266>.
- [9] Lipiäinen S, Lipiäinen K, Ahola A, Vakkilainen E. Use of existing gas infrastructure in European hydrogen economy. *Int J Hydrogen Energy* Sep. 2023;48(80):31317–29. <https://doi.org/10.1016/j.ijhydene.2023.04.283>.
- [10] Raj A, et al. Evaluating hydrogen gas transport in pipelines: current state of numerical and experimental methodologies. *Int J Hydrogen Energy* May 2024;67:136–49. <https://doi.org/10.1016/j.ijhydene.2024.04.140>.
- [11] Dinh QV, Pereira PHT, Dinh VN, Nagle AJ, Leahy PG. Levelised cost of transmission comparison for green hydrogen and ammonia in new-build offshore energy infrastructure: pipelines, tankers, and HVDC. *Int J Hydrogen Energy* Apr. 2024;62:684–98. <https://doi.org/10.1016/j.ijhydene.2024.03.066>.
- [12] Cerniauskas S, Jose Chavez Junco A, Grube T, Robinus M, Stolten D. Options of natural gas pipeline reassignment for hydrogen: cost assessment for a Germany case study. *Int J Hydrogen Energy* Apr. 2020;45(21):12095–107. <https://doi.org/10.1016/j.ijhydene.2020.02.121>.
- [13] Nicolle A, Massol O. Build more and regret less: oversizing H₂ and CCS pipeline systems under uncertainty. *Energy Pol* Aug. 2023;179:113625. <https://doi.org/10.1016/j.enpol.2023.113625>.
- [14] Bell IH, Wronski J, Quoilin S, Lemort V. Pure and pseudo-pure fluid thermophysical property evaluation and the open-source thermophysical property library CoolProp. *Ind Eng Chem Res* Feb. 2014;53(6):2498–508. <https://doi.org/10.1021/ie4033999>.
- [15] The Engineering Toolbox. The engineering ToolBox (2003). Fuels - higher and lower calorific values [online]. Available at: https://www.engineeringtoolbox.com/fuels-higher-calorific-values-d_169.html; 2003.

- [16] Lauermann G, Häussinger P, Lohmüller R, Watson AM. Hydrogen, 1. Properties and occurrence. In: Ullmann's encyclopedia of industrial chemistry, Wiley-VCH Verlag GmbH & Co. KGaA. Weinheim, Germany: Wiley-VCH Verlag GmbH & Co. KGaA; 2013. p. 1–15. <https://doi.org/10.1002/14356007.a13.297.pub3>.
- [17] Dodds PE, Demoullin S. Conversion of the UK gas system to transport hydrogen. *Int J Hydrogen Energy* Jun. 2013;38(18):7189–200. <https://doi.org/10.1016/j.ijhydene.2013.03.070>.
- [18] Quarton CJ, Samsatli S. Should we inject hydrogen into gas grids? Practicalities and whole-system value chain optimisation. *Appl Energy* Oct. 2020;275:115172. <https://doi.org/10.1016/j.apenergy.2020.115172>.
- [19] Wachsmuth J, et al. Roadmap Gas für die Energiewende - Nachhaltiger Klimabeitrag des Gassektors. Umweltbundesamt; 2019 [Online]. Available: <http://www.umweltbundesamt.de/publikationen/roadmap-gas-fuer-die-energiawende-e-nachhaltiger>. [Accessed 3 February 2023].
- [20] Weber M, Perrin J. "Hydrogen transport and distribution," in hydrogen technology. In: Léon A, editor. Green energy and technology; 2008. p. 129–49. https://doi.org/10.1007/978-3-540-69925-5_4. Berlin, Heidelberg: Springer Berlin Heidelberg.
- [21] EGIG. Gas pipeline incidents: 11th report of the European gas pipeline incident data Group (period 1970 - 2019). EGIG; 2020 [Online]. Available: <https://www.egig.eu/reports>. [Accessed 3 February 2023].
- [22] Ríos-Mercado RZ, Borraz-Sánchez C. Optimization problems in natural gas transportation systems: a state-of-the-art review. *Appl Energy* Jun. 2015;147:536–55. <https://doi.org/10.1016/j.apenergy.2015.03.017>.
- [23] Abbas AJ, Hassani H, Burby M, John IJ. An investigation into the volumetric flow rate requirement of hydrogen transportation in existing natural gas pipelines and its safety implications. *Gas Oct.* 2021;1(4):156–79. <https://doi.org/10.3390/gases1040013>.
- [24] Mischner J, Fasold H-G. Gasbeschaffenheiten in Deutschland: was zum Wobbe-Index noch gesagt werden sollte. 2019.
- [25] Galyas AB, Kis L, Tihanyi L, Szunyog I, Vadaszi M, Koncz A. Effect of hydrogen blending on the energy capacity of natural gas transmission networks. *Int J Hydrogen Energy* Jan. 2023;50360319922059146. <https://doi.org/10.1016/j.ijhydene.2022.12.198>.
- [26] Faye O, Szpunar J, Eduok U. A critical review on the current technologies for the generation, storage, and transportation of hydrogen. *Int J Hydrogen Energy* Apr. 2022;47(29):13771–802. <https://doi.org/10.1016/j.ijhydene.2022.02.112>.
- [27] van der Zwaan BCC, Schoots K, Rivera-Tinoco R, Verbong GPJ. The cost of pipelining climate change mitigation: an overview of the economics of CH₄, CO₂ and H₂ transportation. *Appl Energy* Nov. 2011;88(11):3821–31. <https://doi.org/10.1016/j.apenergy.2011.05.019>.
- [28] Gondal IA. Hydrogen transportation by pipelines. In: Compendium of hydrogen energy. Elsevier; 2016. p. 301–22. <https://doi.org/10.1016/B978-1-78242-362-1.00012-2>.
- [29] Kopasz J. Fuel cells and odorants for hydrogen. *Int J Hydrogen Energy* Sep. 2007;32(13):2527–31. <https://doi.org/10.1016/j.ijhydene.2006.11.001>.
- [30] Stöhr T, Reiter V, Scheikl S, Klopčič N, Brandstätter S, Trattner A. Hydrogen quality in used natural gas pipelines: an experimental investigation of contaminants according to ISO 14687:2019 standard. *Int J Hydrogen Energy* May 2024;67:1136–47. <https://doi.org/10.1016/j.ijhydene.2023.09.305>.
- [31] Bard J, Gerhardt N, Selzam P, Beil M, Wiemer DM, Buddensiek M. The limitations of hydrogen blending in the European gas grid - a study on the use, limitations and cost of hydrogen blending in the European gas grid at the transport and distribution level. Fraunhofer IEE; 2022.
- [32] Neugebauer R, editor. Wasserstofftechnologien. Berlin, Heidelberg: Springer Berlin Heidelberg; 2022. <https://doi.org/10.1007/978-3-662-64939-8>.
- [33] Hollingsworth J, Phillippi G, Hinchliff M, Kulhanek C, Rimpel AM, Maywald F. Reciprocating compressors. In: Compression machinery for oil and gas. Elsevier; 2019. p. 167–252. <https://doi.org/10.1016/B978-0-12-814683-5.00005-5>.
- [34] Witkowski A, Rusin A, Majkut M, Stolecka K. Analysis of compression and transport of the methane/hydrogen mixture in existing natural gas pipelines. *Int J Pres Ves Pip Sep.* 2018;166:24–34. <https://doi.org/10.1016/j.iijpvp.2018.08.002>.
- [35] Häussinger P, Lohmüller R, Watson AM. Hydrogen, 5. Handling. In: Ullmann's encyclopedia of industrial chemistry. John Wiley & Sons, Ltd; 2011. <https://doi.org/10.1002/14356007.o13.o06>.
- [36] Witkowski A, Rusin A, Majkut M, Stolecka K. Comprehensive analysis of hydrogen compression and pipeline transportation from thermodynamics and safety aspects. *Energy* Dec. 2017;141:2508–18. <https://doi.org/10.1016/j.energy.2017.05.141>.
- [37] White B, Kreuz T, Simons S. Midstream. In: Compression machinery for oil and gas. Elsevier; 2019. p. 387–400. <https://doi.org/10.1016/B978-0-12-814683-5.00009-2>.
- [38] Folga SM. Natural gas pipeline technology overview. ANL/EVS/TM/08-5 Nov. 2007;925391. <https://doi.org/10.2172/925391>.
- [39] Guandalini G, Colbertaldo P, Campanari S. Dynamic modeling of natural gas quality within transport pipelines in presence of hydrogen injections. *Appl Energy* Jan. 2017;185:1712–23. <https://doi.org/10.1016/j.apenergy.2016.03.006>.
- [40] White B, Kreuz T, Talabisco G. Equipment overview. In: Compression machinery for oil and gas. Elsevier; 2019. p. 13–27. <https://doi.org/10.1016/B978-0-12-814683-5.00002-X>.
- [41] Hammer G, et al. Natural gas. In: Ullmann's encyclopedia of industrial chemistry, Wiley-VCH Verlag GmbH & Co. KGaA. Weinheim, Germany: Wiley-VCH Verlag GmbH & Co. KGaA; 2006. <https://doi.org/10.1002/14356007.a17.073.pub2>. a17_073.pub2.
- [42] dena. Dena-leitstudie aufbruch klimaneutralität. In: Deutsche energie-agentur GmbH (Hrsg.); 2021.
- [43] Prognos, Öko-Institut, Wuppertal-Institut. Klimaneutrales Deutschland. Studie im Auftrag von Agora Energiewende, Agora Verkehrswende und Stiftung Klimaneutralität. Prognos, Öko-Institut, Wuppertal-Institut; 2020.
- [44] Laureys A, Depaetere R, Cauwels M, Depover T, Hertelé S, Verbeke K. Use of existing steel pipeline infrastructure for gaseous hydrogen storage and transport: a review of factors affecting hydrogen induced degradation. *J Nat Gas Sci Eng May* 2022;101:104534. <https://doi.org/10.1016/j.jngse.2022.104534>.
- [45] Barthélémy H. Effects of pressure and purity on the hydrogen embrittlement of steels. *Int J Hydrogen Energy* Feb. 2011;36(3):2750–8. <https://doi.org/10.1016/j.ijhydene.2010.05.029>.
- [46] Holbrook JH, Cialone HJ, Collings EW, Drauglis EJ, Scott PM, Mayfield ME. Control of hydrogen embrittlement of metals by chemical inhibitors and coatings. In: Gaseous hydrogen embrittlement of materials in energy technologies. Elsevier; 2012. p. 129–53. <https://doi.org/10.1533/9780857095374.1.129>.
- [47] Schröder V, Emonts B, Janßen H, Schulze H-P. Explosion limits of hydrogen/oxygen mixtures at initial pressures up to 200 bar. *Chem Eng Technol Aug.* 2004;27(8):847–51. <https://doi.org/10.1002/ceat.200403174>.
- [48] Luo Z, Hu Y, Xu H, Gao D, Li W. Cost-economic analysis of hydrogen for China's fuel cell transportation field. *Energies* Dec. 2020;13(24):6522. <https://doi.org/10.3390/en13246522>.
- [49] Nordin M, Wassie SA, Van Sint Annaland M, Pacheco Tanaka DA, Viviente Sole JL, Gallucci F. Techno-economic evaluation on a hybrid technology for low hydrogen concentration separation and purification from natural gas grid. *Int J Hydrogen Energy* Jul. 2021;46(45):23417–35. <https://doi.org/10.1016/j.ijhydene.2020.05.009>.
- [50] Atrens A, Gray E, Venezuela J, Hoschke J, Roethig M. Feasibility of the use of gas phase inhibition of hydrogen embrittlement in gas transmission pipelines carrying hydrogen: a review. *J Occup Med Jan.* 2023;75(1):232–8. <https://doi.org/10.1007/s11837-022-05559-8>.
- [51] Krieg D. Konzept und Kosten eines Pipelinesystems zur Versorgung des deutschen Straßenverkehrs mit Wasserstoff. In: Schriften des Forschungszentrums Jülich. Reihe Energie und Umwelt/energy and environment. Verlag: Forschungszentrum Jülich GmbH Zentralbibliothek; 2012. no. 144. Jülich.
- [52] Esakul KA, Mason J. Nonmetallic applications in oil and gas production (pipes, liners, rehabilitations). In: Trends in oil and gas corrosion research and technologies. Elsevier; 2017. p. 627–60. <https://doi.org/10.1016/B978-0-08-101105-8.00027-9>.
- [53] Zeng D, Deng K, Lin Y, Shi T, Shi D, Zhou L. Theoretical and experimental study of the thermal strength of anticorrosive lined steel pipes. *Petrol Sci Sep.* 2014;11(3):417–23. <https://doi.org/10.1007/s12182-014-0356-z>.
- [54] Byrnes T. Pipeline coatings. In: Trends in oil and gas corrosion research and technologies. Elsevier; 2017. p. 563–91. <https://doi.org/10.1016/B978-0-08-101105-8.00024-3>.
- [55] Kehr JA, Enos DG. FBE, a foundation for pipeline corrosion coatings, Paper presented at the CORROSION 2000. Mar. 2000.
- [56] Menon ES. Gas pipeline hydraulics. 0 ed. CRC Press; 2005. <https://doi.org/10.1201/9781420038224>.
- [57] Hanke R, Holm T, Ulleberg Ø. The case for high-pressure PEM water electrolysis. *Energy Convers Manag Jun.* 2022;261:115642. <https://doi.org/10.1016/j.enconman.2022.115642>.
- [58] Kost C, Shammugam S, Fluri V, Peper D, Memar AD, Schlegel T. Levelized cost of electricity renewable energy technologies. Fraunhofer Institute for solar energy systems ISE; 2021.
- [59] Kreidelmeyer S, Dambeck H, Kirchner Dr A, Wünsch M. Kosten und Transformationspfade für strombasierte Energieträger - Endbericht zum Projekt. In: Transformationspfade und regulatorischer Rahmen für synthetische Brennstoffe. Prognos AG; 2020.
- [60] IEA. Global hydrogen review 2023. Paris: IEA; 2023 [Online]. Available: <https://www.iea.org/reports/global-hydrogen-review-2023>.
- [61] Mendibide C, Vucko F, Martinez M, Joshi GR, Kittel J. Effect of degraded environmental conditions on the service behavior of a X65 pipeline steel not designed for hydrogen transport. *Int J Hydrogen Energy Jan.* 2024;52:1019–32. <https://doi.org/10.1016/j.ijhydene.2023.05.309>.